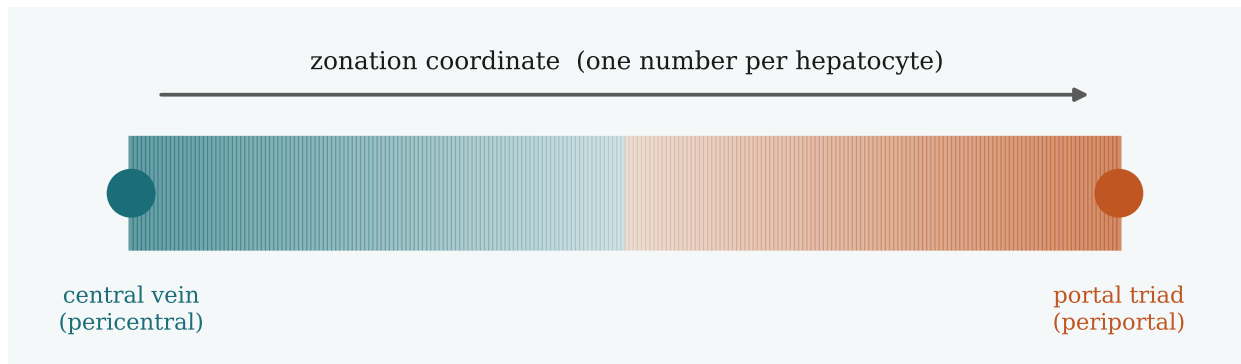


Spatial Hepatocyte Zonation in MASLD Progression

what changes, what doesn't, and how sure we are

A confounder-controlled, mechanism-resolved analysis.



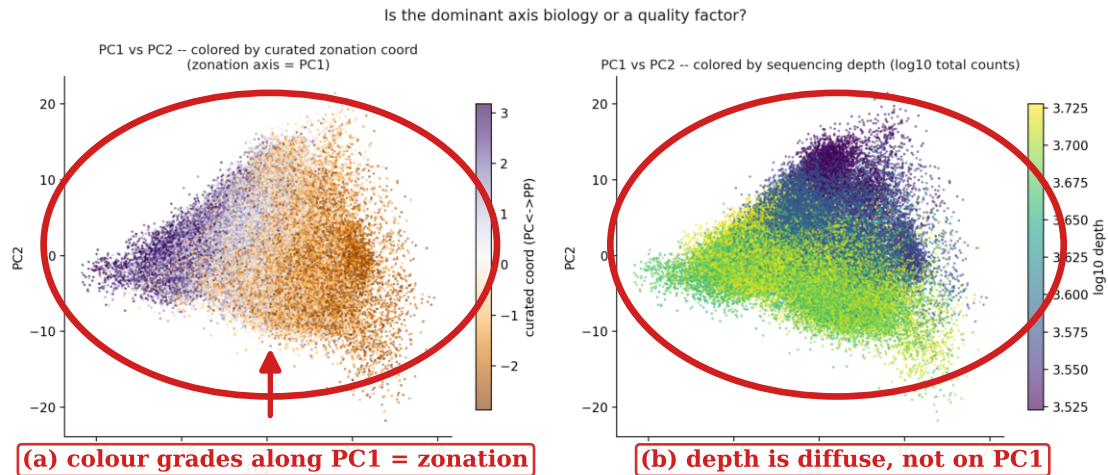
The healthy liver is zoned: hepatocytes near the central vein (pericentral) and near the portal triad (periportal) run opposite metabolic programs along a single porto-central axis. We built TWO co-primary rulers to read that axis — a curated gene-set ruler and a label-free PCA axis — validated both on healthy tissue, then transferred them unchanged to disease. Every readout in this report uses the donor as the unit of inference (47 donors, ~69,426 hepatocytes; ruler = `expanded_curated` unless noted).

Methods caveat, stated up front

The scored matrix is the SCT-corrected assay (depth-regularized; per-cell totals ~3–5.7k), so sequencing depth is already largely controlled at the source, before any of the confounder analyses below. Section 4 shows the result survives explicit depth equalization on top of this.

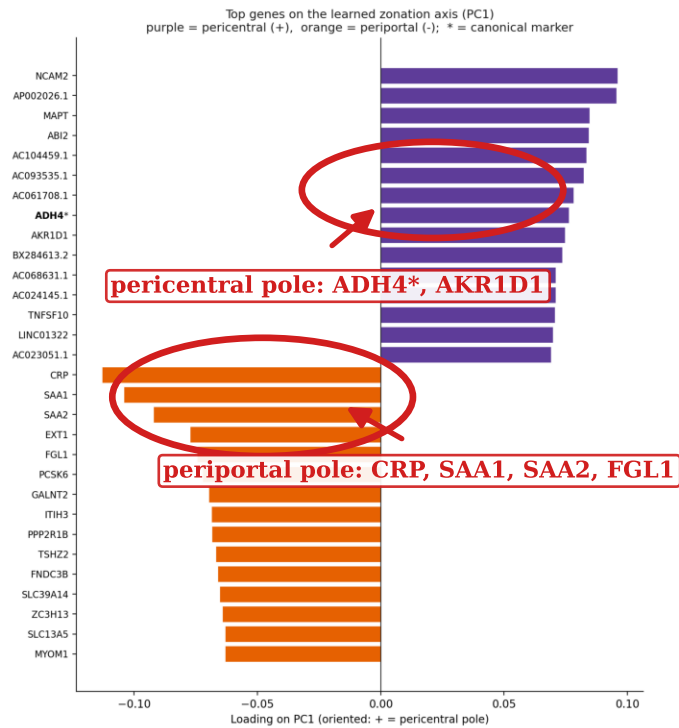
1. The ruler is real biology

Before trusting an unsupervised axis, we must show it is zonation and not a technical artifact. The dominant axis of variation (PC1) IS the zonation axis: it correlates $|r| = 0.76$ with the curated zonation coordinate and $|r| = 0.54$ with the canonical 8-marker axis, and it carries 5.8% of total variance — about twice the next component. Sequencing depth and quality load on PC2, NOT on PC1.



e_pca_pc12_map: PC1 vs PC2 coloured by the curated zonation coordinate (left) and by sequencing depth (right). The smooth left→right colour gradient in (a) means PC1 IS the zonation axis; in (b) depth is scattered with no PC1 alignment.

The loadings read as genuine biology, not noise. The periportal pole is led by acute-phase and fibrinogen-like genes (CRP, SAA1, SAA2, FGL1); the pericentral pole is led by bile-acid / alcohol metabolism genes (ADH4, AKR1D1). These are exactly the textbook zonal programs.



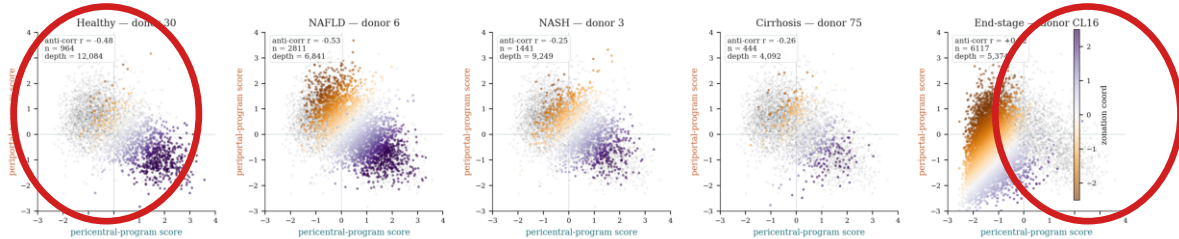
e_pca_loadings: top genes on the learned PC1 axis. Purple = pericentral (+) pole, orange = periportal (-) pole; * marks a canonical marker. The poles recover known zonal biology.

2. Healthy zonation is clear — and collapses with disease

With a validated ruler in hand, the first disease question is whether the pericentral and periportal programs stay anti-correlated within a donor (as they must if zonation is intact). Per-donor PC-vs-PP scatters answer it directly. In Healthy tissue the cloud is a tight anti-diagonal (well-powered donor 30, $r = -0.48$), and it stays anti-correlated through NAFLD/NASH ($r = -0.53, -0.41$). By Cirrhosis and End-stage the anti-diagonal is gone — the cloud becomes a positive blob ($r = +0.26, +0.22$): the two programs now rise and fall together, the signature of a collapsed axis.

Healthy: tight anti-diagonal

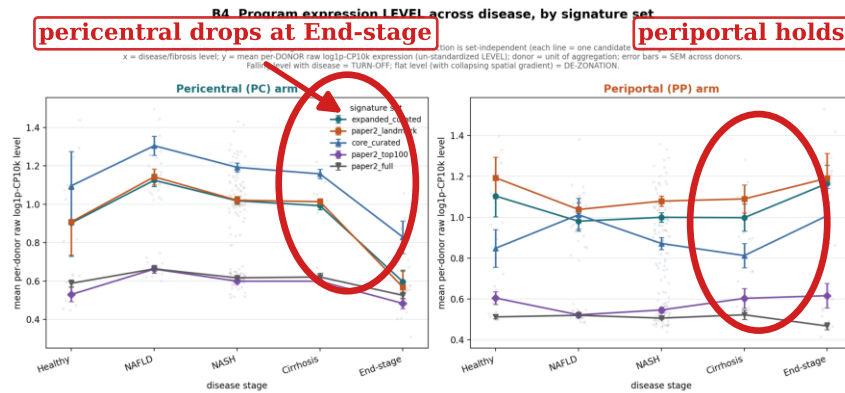
End-stage: positive blob



a1_pcpp_representatives_stage: per-donor pericentral vs periportal program scores, one representative donor per stage (Healthy→End-stage). The within-donor anti-diagonal (left) gives way to a positive blob (right).

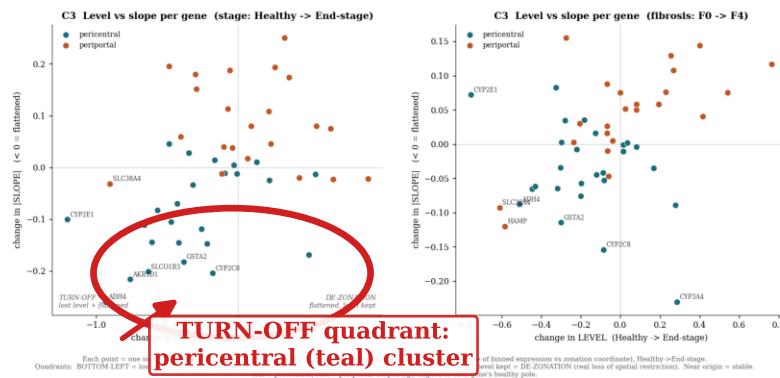
3. The mechanism is asymmetric and specific

A collapsed axis could happen two ways: the spatial GRADIENT could flatten (de-zonation), or one program could simply switch OFF (turn-off). These have different biology, so we separate them. The answer is clear: pericentral TURN-OFF dominates. Pericentral program expression LEVEL drops ~34-37% by End-stage, and every gene set agrees on the direction.



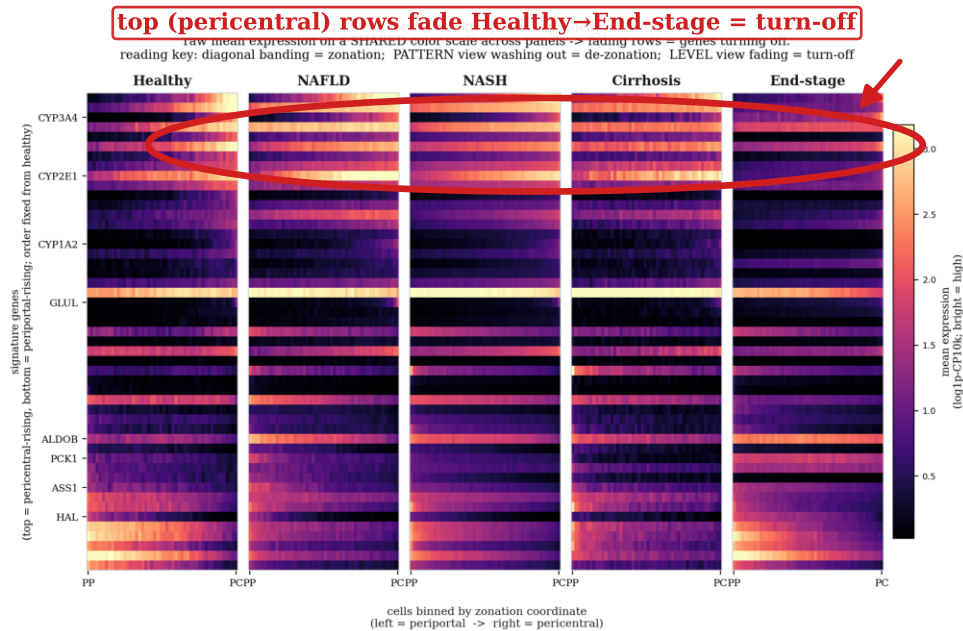
b4_set_expression_by_stage: mean per-donor program level by stage for every signature set. Left (pericentral) arms fall at End-stage; right (periportal) arms stay flat — the loss is asymmetric.

Gene by gene, 14/26 pericentral genes lose BOTH level and gradient (the turn-off quadrant; e.g. CYP2E1 dLevel -1.21, ADH4 -0.93), while 22/23 periportal genes hold their level (stable). The level-vs-slope map makes the asymmetry visible: pericentral (teal) genes cluster in the bottom-left turn-off quadrant.

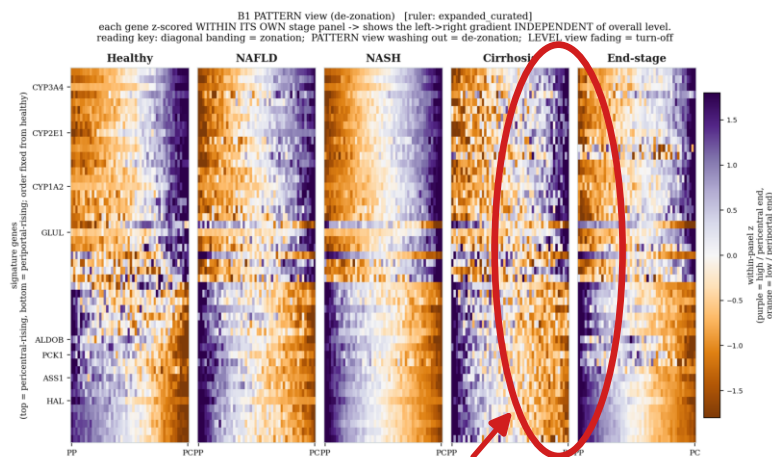


c3_level_vs_slope (stage panel): each point is one signature gene; x = change in LEVEL, y = change in gradient [slope], Healthy→End-stage. Pericentral genes (teal) pile into the bottom-left TURN-OFF quadrant (lost level AND slope).

Now the key dissociation. The pericentral heatmap ROWS fade in the LEVEL view (genes turning off), but in the PATTERN view — where each gene is z-scored within its own panel, removing level — the diagonal banding largely PERSISTS to Cirrhosis. So pure gradient loss (de-zonation) is the WEAKER, secondary effect; turn-off is primary. The bipolar-axis collapse seen in Section 2 is most likely downstream of pericentral silencing.



b1_heatmap_level_stage: raw mean expression on a shared colour scale across stages (left=Healthy → right=End-stage). The top pericentral rows (CYP3A4/CYP2E1...) fade rightward — the genes turn off.

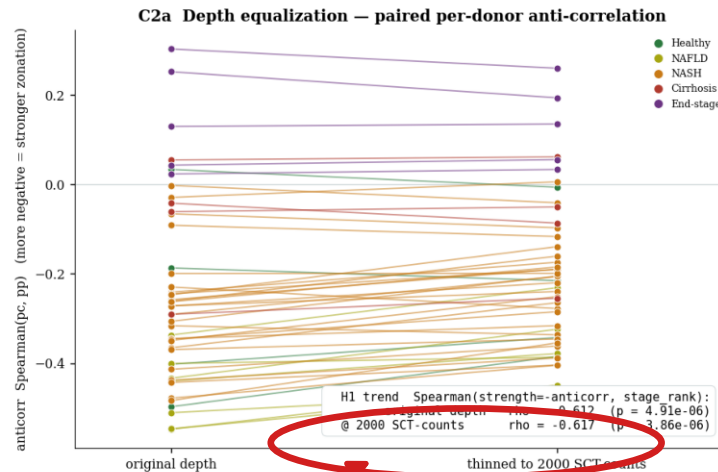


banding PERSISTS to Cirrhosis = pattern survives

b1_heatmap_pattern_stage: each gene z-scored WITHIN its own panel (level removed). The diagonal banding — the spatial gradient — largely survives into Cirrhosis, so de-zonation is the weaker effect.

4. It is robust — confounders controlled

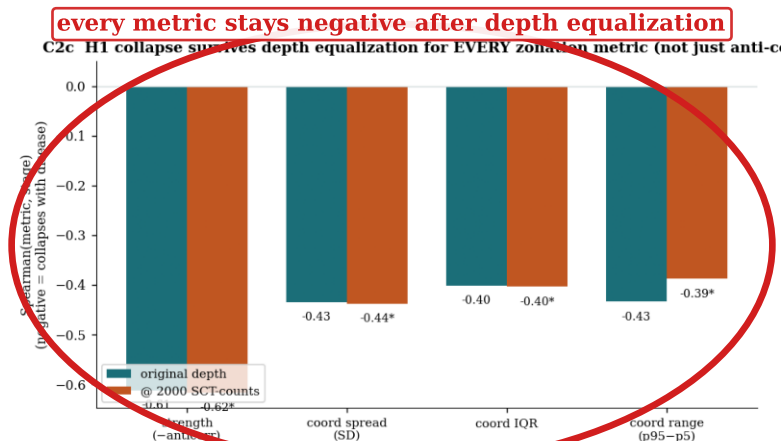
A reviewer's first worry is that the collapse is a sampling or depth artifact. It is not. The headline trend — Spearman(zonation strength, stage) = -0.612 — is essentially unmoved by equalizing cell count across donors (common-N = 200 → -0.586) and by thinning EVERY cell to a common depth and re-scoring (-0.617).



trend survives depth equalization: $-0.612 \rightarrow -0.617$

c2_depth_paired: per-donor anti-correlation at original depth vs thinned to a common 2000 SCT-counts. Deep, strongly-zonated donors weaken toward 0 when thinned, but the STAGE trend (boxed) is unchanged: $-0.612 \rightarrow -0.617$.

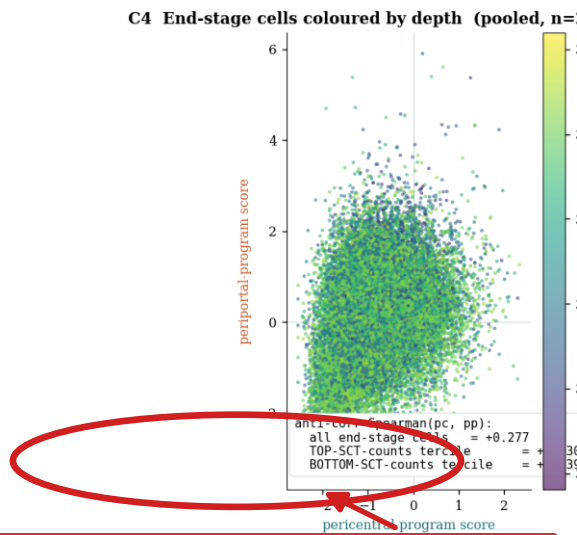
And this is not specific to the anti-correlation readout. We repeated the depth-equalization test for EVERY zonation metric — coordinate spread (SD), inter-quartile range, and 5-95% range — and all four keep a significant collapsing trend at common depth (strength -0.62 , spread -0.44 , IQR -0.40 , range -0.39 ; all $p < 0.05$). The collapse is a property of the data, not of one metric or of sequencing depth.



c2_multimetric_survival: Spearman(metric, stage) at original depth (teal) vs common depth (rust) for four independent zonation readouts. All stay negative (collapsing) and significant — the depth control is not anti-correlation-specific.

Depth DOES causally degrade the measurement — a deep, zonated donor weakens from $r = -0.55$ to -0.33 when thinned — but depth does not track disease stage, so it cannot manufacture the trend. The End-stage positive blob is likewise NOT a depth artifact: among End-stage cells, the BEST-sequenced ones are if anything MORE positive (high-depth tercile $+0.33$ vs low-depth tercile $+0.24$), the opposite of what a depth artifact

would predict.

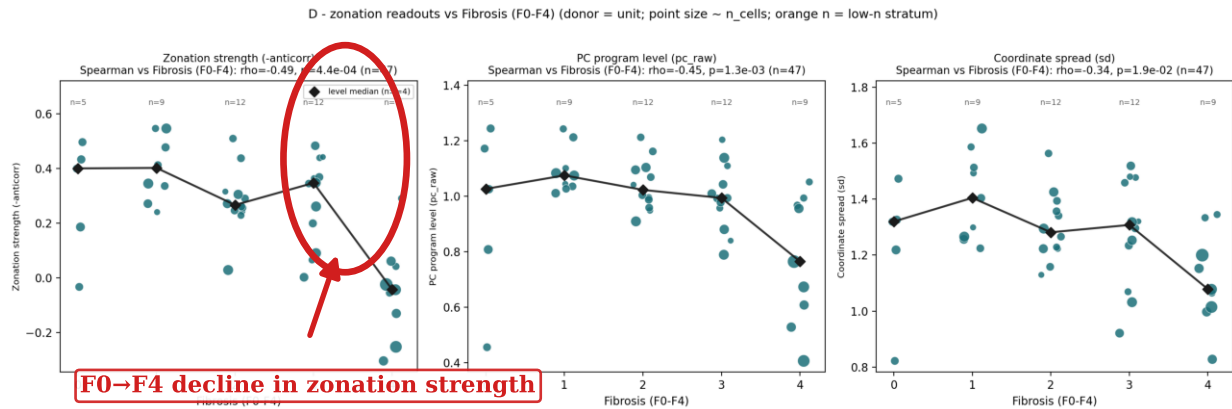


best-sequenced cells are MORE positive, not less

c4_endstage_umi_scatter: End-stage cells coloured by depth. The anti-correlation is positive overall (+0.277) and the high-depth tercile is MORE positive than the low — so high-depth cells are not more anti-correlated. The collapse is not explained by depth.

5. It tracks fibrosis specifically

Finally, the de-zonation signal is not generic disease severity — it is tied to FIBROSIS. Zonation strength declines monotonically across the fibrosis stages F0→F4 (Spearman $\rho = -0.49$, $p = 4.4e-4$, $n = 47$). And it is fibrosis-SPECIFIC: the partial correlation controlling for the NAS activity score holds at $\rho = -0.40$, whereas the reverse — strength vs NAS controlling for fibrosis — collapses to -0.12 .



d_zonation_vs_fibrosis: donor-level zonation readouts vs Fibrosis (F0-F4); point size ~ n_{cells} .
The leftmost panel (zonation strength) declines from F0/F1 to F4 ($\rho = -0.49$).

6. Honest caveats

These limitations are stated plainly so the strength of each claim is not over-read.

- Only 4 healthy-control donors anchor the “healthy” end, and 2 of them are low-depth and weakly zoned — the healthy baseline is thinly powered.
- De-zonation (pure gradient loss) is the WEAKER effect; the dominant, primary mechanism is pericentral turn-off. The two should not be conflated.
- Within “NASH-without-cirrhosis” the F1→F3 grading is essentially flat: the fibrosis signal is the F0/F1-vs-F4 contrast, not fine intermediate grading.
- The depth control operates on the SCT-corrected scale, not raw UMI counts — it removes residual depth effects on top of SCT, not raw-count depth from scratch.
- This is a cross-sectional human cohort. We therefore establish a confounder-ROBUST ASSOCIATION between fibrosis and de-zonation — NOT disease→de-zonation causality.

What the evidence does and does not support

Supported: (i) PC1 is genuine zonation biology; (ii) the bipolar axis collapses with disease; (iii) the mechanism is asymmetric pericentral turn-off, with de-zonation secondary; (iv) the trend is robust to cell-count and depth equalization; (v) it tracks fibrosis specifically. Not supported: a causal claim, fine within-NASH grading, or a strong healthy baseline.

7. Methods recap

Rulers (co-primary). (1) Curated signature-mean coordinate: mean_z(pericentral) – mean_z(periportal) over per-gene z-scores. (2) A label-free PCA axis (PC1) learned without any gene list. Both must pass an 8-marker healthy validation gate before transfer to disease. Readouts span four figure families: A (axis collapse), B (mechanism: level vs pattern), C (confounder controls: cell count, depth, End-stage depth), D (fibrosis/NAS staging), and E (ruler interpretation: PCA map and loadings) — ~25 figures in all. All inference is donor-level across ~47 donors (~69,426 hepatocytes), with the donor as the unit throughout.